

University of Makati
College of Computing and Information Sciences

ELEC 5 – Group Activity

Members:

Ramos, Anne Lorraine F
Dayit, Jessllyn S.
Montealegre, Ezra David L

Scenario: Latte Runners Vince's Vinyl (A Music Record Shop)

Vince is eager to get going. Just today he had a customer come in and sell him a dozen old albums. One is quite rare and could be worth a lot of money. Vince doesn't want to lose track of it. He is ready to get organized and start entering his transactions in the database. You review your design with him and promise that you will begin building the database immediately. But, you remind him, it is important to test the database before actually starting to use it for the business.

1. Review your diagram for the database making sure that the design is complete and normalized.
2. Create the database in SQL Server

```
mysql> CREATE DATABASE VinceVinylShop;
Query OK, 1 row affected (0.00 sec)

mysql> USE VinceVinylShop;
Database changed
```

3. Create the tables in the new database, selecting appropriate data types for the columns, setting a primary key for each table, and setting allow nulls as appropriate

```
mysql> CREATE TABLE Customers (
->     CustomerID INT PRIMARY KEY AUTO_INCREMENT,
->     FirstName VARCHAR(50) NOT NULL,
->     LastName VARCHAR(50) NOT NULL,
->     Email VARCHAR(100) NOT NULL,
->     PhoneNumber VARCHAR(20) NOT NULL,
->     DateJoined DATE NOT NULL
-> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Albums (
->     AlbumID INT PRIMARY KEY AUTO_INCREMENT,
->     Title VARCHAR(100) NOT NULL,
->     Artist VARCHAR(100) NOT NULL,
->     Genre VARCHAR(50) NOT NULL,
->     ReleaseYear INT NOT NULL,
->     Price DECIMAL(10, 2) NOT NULL,
->     IsRare BIT NOT NULL,
->     QuantityInStock INT NOT NULL
-> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Employees (  
->     EmployeeID INT PRIMARY KEY AUTO_INCREMENT,  
->     FirstName VARCHAR(50) NOT NULL,  
->     LastName VARCHAR(50) NOT NULL,  
->     JobTitle VARCHAR(100) NOT NULL,  
->     HireDate DATE NOT NULL,  
->     Email VARCHAR(100) NOT NULL,  
->     PhoneNumber VARCHAR(20) NOT NULL  
-> );  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE Sales (  
->     SaleID INT PRIMARY KEY AUTO_INCREMENT,  
->     CustomerID INT NOT NULL,  
->     EmployeeID INT NOT NULL,  
->     TransactionDate DATE NOT NULL,  
->     TotalAmount DECIMAL(10, 2) NOT NULL,  
->     FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID),  
->     FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE Suppliers (  
->     SupplierID INT PRIMARY KEY AUTO_INCREMENT,  
->     SupplierName VARCHAR(100) NOT NULL,  
->     ContactPerson VARCHAR(50) NOT NULL,  
->     Email VARCHAR(100) NOT NULL,  
->     PhoneNumber VARCHAR(20) NOT NULL  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

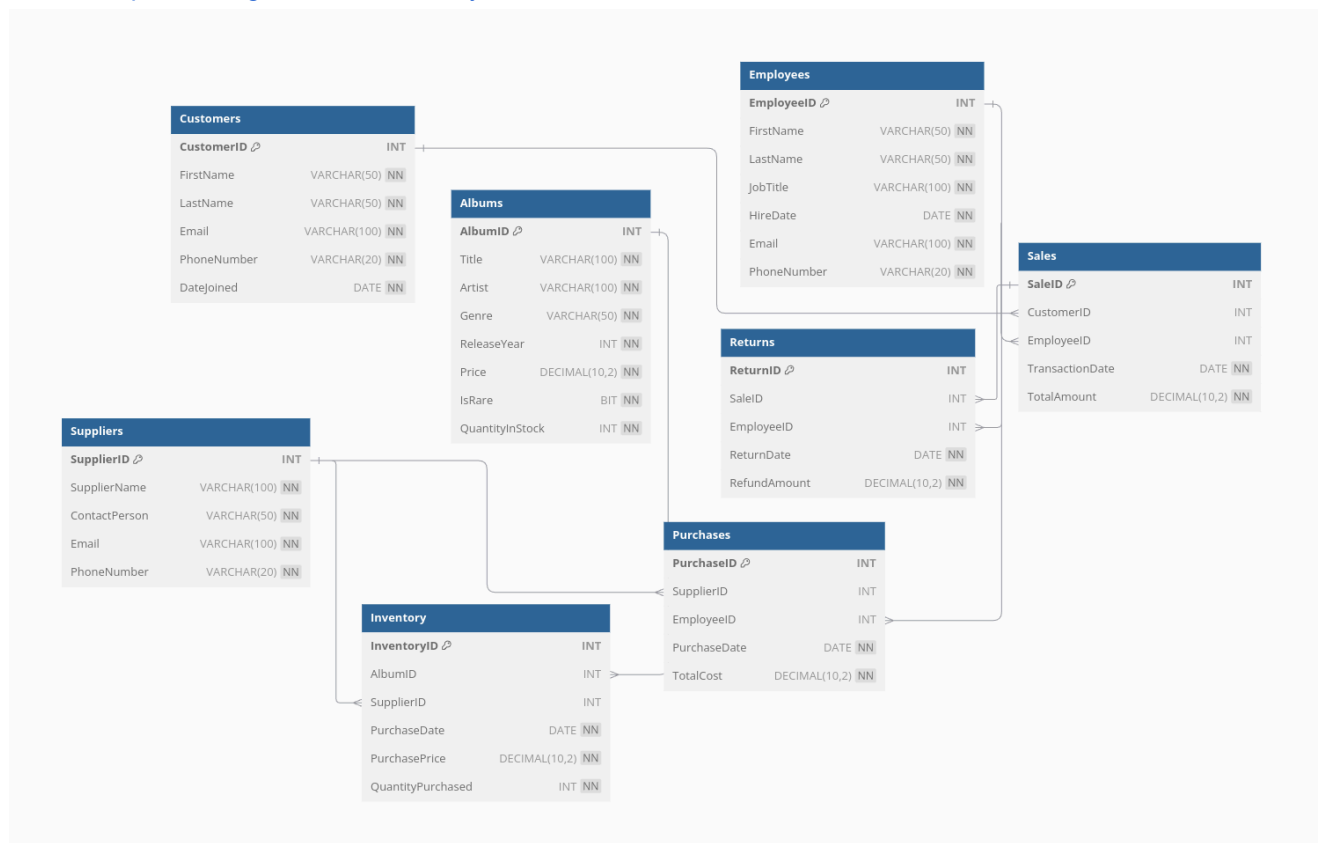
```
mysql> CREATE TABLE Inventory (  
->     InventoryID INT PRIMARY KEY AUTO_INCREMENT,  
->     AlbumID INT NOT NULL,  
->     SupplierID INT NOT NULL,  
->     PurchaseDate DATE NOT NULL,  
->     PurchasePrice DECIMAL(10, 2) NOT NULL,  
->     QuantityPurchased INT NOT NULL,  
->     FOREIGN KEY (AlbumID) REFERENCES Albums(AlbumID),  
->     FOREIGN KEY (SupplierID) REFERENCES Suppliers(SupplierID)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE Purchases (
->     PurchaseID INT PRIMARY KEY AUTO_INCREMENT,
->     SupplierID INT NOT NULL,
->     EmployeeID INT NOT NULL,
->     PurchaseDate DATE NOT NULL,
->     TotalCost DECIMAL(10, 2) NOT NULL,
->     FOREIGN KEY (SupplierID) REFERENCES Suppliers(SupplierID),
->     FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID)
-> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Returns (
->     ReturnID INT PRIMARY KEY AUTO_INCREMENT,
->     SaleID INT NOT NULL,
->     EmployeeID INT NOT NULL,
->     ReturnDate DATE NOT NULL,
->     RefundAmount DECIMAL(10, 2) NOT NULL,
->     FOREIGN KEY (SaleID) REFERENCES Sales(SaleID),
->     FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID)
-> );
Query OK, 0 rows affected (0.04 sec)
```

4. Create a database diagram and create the relationships among tables
For better viewing of the diagram:

<https://dbdiagram.io/d/VinceVinyl2-66e2dcda6dde7f4149e7a03d>



5. Add some sample data to each table

```
mysql> INSERT INTO Customers (FirstName, LastName, Email, PhoneNumber, DateJoined) VALUES
-> ('Jhon Mark', 'Enrique', 'jhon.enrique@email.com', '09171234567', '2023-08-10'),
-> ('Clyde', 'Canlas', 'clyde.canlas@email.com', '09181234567', '2022-07-22'),
-> ('Jomar', 'Metoda', 'jomar.metoda@email.com', '09191234567', '2023-01-15'),
-> ('Jomari', 'Tumpang', 'jomari.tumpang@email.com', '09201234567', '2023-05-05'),
-> ('Patrick', 'Formanes', 'patrick.formanes@email.com', '09211234567', '2021-03-14'),
-> ('Daniel', 'Vidar', 'daniel.vidar@email.com', '09221234567', '2022-12-30');
Query OK, 6 rows affected (0.01 sec)
Records: 6 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Albums (Title, Artist, Genre, ReleaseYear, Price, IsRare, QuantityInStock) VALUES
-> ('1989', 'Taylor Swift', 'Pop', 2014, 19.99, 0, 50),
-> ('Evermore', 'Taylor Swift', 'Alternative', 2020, 21.99, 0, 40),
-> ('Emails I Can't Send', 'Sabrina Carpenter', 'Pop', 2022, 18.99, 0, 30),
-> ('Singular Act II', 'Sabrina Carpenter', 'Pop', 2019, 17.99, 0, 25),
-> ('LM5', 'Little Mix', 'Pop', 2018, 16.99, 0, 20),
-> ('Abbey Road', 'The Beatles', 'Rock', 1969, 99.99, 1, 5),
-> ('Sgt. Pepper's Lonely Hearts Club Band', 'The Beatles', 'Rock', 1967, 109.99, 1, 3);
Query OK, 7 rows affected (0.00 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Employees (FirstName, LastName, JobTitle, HireDate, Email, PhoneNumber) VALUES
-> ('Anne Lorraine', 'Ramos', 'Manager/Owner', '2020-01-01', 'anne.ramos@email.com', '09181234567'),
-> ('Ezra David', 'Montealegre', 'Sales Associate', '2021-06-15', 'ezra.montealegre@email.com', '09191234567'),
-> ('Jesslhyn', 'Dayrit', 'Inventory Specialist', '2022-03-20', 'jesslhyn.dayrit@email.com', '09201234567');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Suppliers (SupplierName, ContactPerson, Email, PhoneNumber) VALUES
-> ('Big Machine Records', 'Scott Borchetta', 'info@bigmachine.com', '09991234567'),
-> ('Island Records', 'John Janick', 'info@islandrecords.com', '09981234567'),
-> ('Columbia Records', 'Rob Stringer', 'info@columbiarecords.com', '09971234567'),
-> ('Apple Corps', 'Jeff Jones', 'info@applecorps.com', '09961234567');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Sales (CustomerID, EmployeeID, TransactionDate, TotalAmount)
VALUES
  -> (1, 2, '2023-08-12', 19.99), -- Jhon Mark buys '1989'
  -> (2, 1, '2023-07-25', 99.99), -- Clyde buys 'Abbey Road'
  -> (3, 3, '2023-01-17', 18.99), -- Jomar buys 'Emails I Can't Send'
  -> (4, 1, '2023-05-07', 16.99), -- Jomari buys 'LM5'
  -> (5, 3, '2021-03-16', 109.99), -- Patrick buys 'Sgt. Pepper's Lonely Heart
s Club Band'
  -> (6, 2, '2022-12-31', 21.99); -- Daniel buys 'Evermore'
Query OK, 6 rows affected (0.00 sec)
Records: 6 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Inventory (AlbumID, SupplierID, PurchaseDate, PurchasePrice,
QuantityPurchased) VALUES
  -> (1, 1, '2023-06-01', 12.99, 50), -- 1989 by Taylor Swift from Big Machine
Records
  -> (2, 1, '2023-06-01', 14.99, 40), -- Evermore by Taylor Swift from Big Mac
hine Records
  -> (3, 2, '2023-05-01', 11.99, 30), -- Emails I Can't Send by Sabrina Carpen
ter from Island Records
  -> (4, 2, '2023-05-01', 10.99, 25), -- Singular Act II by Sabrina Carpenter
from Island Records
  -> (5, 3, '2023-04-01', 9.99, 20), -- LM5 by Little Mix from Columbia Recor
ds
  -> (6, 4, '2023-03-01', 60.00, 5), -- Abbey Road by The Beatles from Apple
Corps
  -> (7, 4, '2023-03-01', 70.00, 3); -- Sgt. Pepper's Lonely Hearts Club Band
by The Beatles from Apple Corps
Query OK, 7 rows affected (0.00 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

```
mysql> ^MINsert INTO Purchases (SupplierID, EmployeeID, PurchaseDate, TotalCost)
VALUES
  -> (1, 3, '2023-06-01', 649.50), -- Big Machine Records purchase by Jesslhyn
Dayrit
  -> (2, 3, '2023-05-01', 479.75), -- Island Records purchase by Jesslhyn Dayr
it
  -> (3, 3, '2023-04-01', 199.80), -- Columbia Records purchase by Jesslhyn Da
yrit
  -> (4, 3, '2023-03-01', 420.00); -- Apple Corps purchase by Jesslhyn Dayrit
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Returns (SaleID, EmployeeID, ReturnDate, RefundAmount) VALUES
  -> (6, 2, '2023-01-05', 21.99); -- Daniel returns 'Evermore'
Query OK, 1 row affected (0.00 sec)
```

6. **Documentation:** Make a Data Dictionary that lists each table, all the columns for that table, the data types for each column.

Customers Table

Column Name	Data Type	Constraints	Description
CustomerID	INT	PRIMARY KEY AUTO_INCREMENT	Unique identifier for each customer.
FirstName	VARCHAR(50)	NOT NULL	Customer's first name.

LastName	VARCHAR(50)	NOT NULL	Customer's last name.
Email	VARCHAR(100)	NOT NULL	Customer's email address.
PhoneNumber	VARCHAR(20)	NOT NULL	Customer's phone number.
DateJoined	DATE	NOT NULL	Date when the customer joined.

Albums Table

Column Name	Data Type	Constraints	Description
AlbumID	INT	PRIMARY KEY AUTO_INCREMENT	Unique identifier for each album.
Title	VARCHAR(100)	NOT NULL	Title of the album.
Artist	VARCHAR(100)	NOT NULL	Artist who performed the album.
Genre	VARCHAR(50)	NOT NULL	Genre of the album.
ReleaseYear	INT	NOT NULL	Year the album was released.
Price	DECIMAL(10, 2)	NOT NULL	Price of the album.
IsRare	BIT	NOT NULL	Indicates if the album is rare (1 for rare, 0 for not).
QuantityInStock	INT	NOT NULL	Number of copies available in stock.

Employees Table

Column Name	Data Type	Constraints	Description
EmployeeID	INT	PRIMARY KEY AUTO_INCREMENT	Unique identifier for each employee.
FirstName	VARCHAR(50)	NOT NULL	Employee's first name.
LastName	VARCHAR(50)	NOT NULL	Employee's last name.
JobTitle	VARCHAR(100)	NOT NULL	Job title of the employee.

HireDate	DATE	NOT NULL	Date when the employee was hired.
Email	VARCHAR(100)	NOT NULL	Employee's email address.
PhoneNumber	VARCHAR(20)	NOT NULL	Employee's phone number.

Sales Table

Column Name	Data Type	Constraints	Description
SaleID	INT	PRIMARY KEY AUTO_INCREMENT	Unique identifier for each sale.
CustomerID	INT	NOT NULL, FOREIGN KEY	References CustomerID in the Customers table.
EmployeeID	INT	NOT NULL, FOREIGN KEY	References EmployeeID in the Employees table.
TransactionDate	DATE	NOT NULL	Date when the sale occurred.
TotalAmount	DECIMAL(10, 2)	NOT NULL	Total amount of the sale.

Suppliers Table

Column Name	Data Type	Constraints	Description
SupplierID	INT	PRIMARY KEY AUTO_INCREMENT	Unique identifier for each supplier.
SupplierName	VARCHAR(100)	NOT NULL	Name of the supplier.
ContactPerson	VARCHAR(50)	NOT NULL	Name of the contact person at the supplier.
Email	VARCHAR(100)	NOT NULL	Supplier's email address.
PhoneNumber	VARCHAR(20)	NOT NULL	Supplier's phone number.

Inventory Table

Column Name	Data Type	Constraints	Description
-------------	-----------	-------------	-------------

InventoryID	INT	PRIMARY KEY AUTO_INCREMENT	Unique identifier for each inventory record.
AlbumID	INT	NOT NULL, FOREIGN KEY	References AlbumID in the Albums table.
SupplierID	INT	NOT NULL, FOREIGN KEY	References SupplierID in the Suppliers table.
PurchaseDate	DATE	NOT NULL	Date when the album was purchased.
PurchasePrice	DECIMAL(10, 2)	NOT NULL	Price at which the album was purchased.
QuantityPurchased	INT	NOT NULL	Quantity of the album purchased.

Purchase Table

Column Name	Data Type	Constraints	Description
PurchaseID	INT	PRIMARY KEY AUTO_INCREMENT	Unique identifier for each purchase.
SupplierID	INT	NOT NULL, FOREIGN KEY	References SupplierID in the Suppliers table.
EmployeeID	INT	NOT NULL, FOREIGN KEY	References EmployeeID in the Employees table.
PurchaseDate	DATE	NOT NULL	Date when the purchase was made.
TotalCost	DECIMAL(10, 2)	NOT NULL	Total cost of the purchase.

Returns Table

Column Name	Data Type	Constraints	Description
ReturnID	INT	PRIMARY KEY AUTO_INCREMENT	Unique identifier for each return.
SaleID	INT	NOT NULL, FOREIGN KEY	References SaleID in the Sales table.

EmployeeID	INT	NOT NULL, FOREIGN KEY	References EmployeeID in the Employees table.
ReturnDate	DATE	NOT NULL	Date when the return was processed.
RefundAmount	DECIMAL(10, 2)	NOT NULL	Amount refunded to the customer.